Quiz 1

Subject: Statistics2 Statistics2 16 Jul 26 Total Marks: 40.00 Questions: 24 With Answers

QUESTIONS

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Q1 MCQ [0.00 marks]

Formula file

Discrete random variables:

Distribution	PMF ($f_X(k)$)	CDF ($F_X(x)$)	$E[X]$	$\mathrm{Var}(X)$
Uniform(A) $A = \{a, a+1, \ldots, b\}$	$\frac{1}{n}, \quad x = k$ $n = b - a + 1$ $k = a, a+1, \ldots, b$	$\begin{cases} 0 & x < 0 \\ \frac{k-a+1}{n} & k \le x < k+1 \\ & k = a, a+1, \ldots, b-1, b \\ 1 & x \ge n \end{cases}$	$\frac{a+b}{2}$	$\frac{n^2-1}{12}$
Bernoulli(p)	$\begin{cases} p & x = 1 \\ 1-p & x = 0 \end{cases}$	$\begin{cases} 0 & x < 0 \\ 1-p & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$	p	$p(1-p)$
Binomial(n, p)	${}^nC_k p^k (1-p)^{n-k}$, $k = 0, 1, \ldots, n$	$\begin{cases} 0 & x < 0 \\ \sum_{i=0}^{k} {}^nC_i p^i (1-p)^{n-i} & k \le x < k+1 \\ & k = 0, 1, \ldots, n \\ 1 & x \ge n \end{cases}$	np	$np(1-p)$
Geometric(p)	$(1-p)^{k-1} p$, $k = 1, \ldots, \infty$	$\begin{cases} 0 & x < 0 \\ 1-(1-p)^k & k \le x < k+1 \\ & k = 1, \ldots, \infty \end{cases}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$\frac{e^{-\lambda}\lambda^k}{k!}$, $k = 0, 1, \ldots, \infty$	$\begin{cases} 0 & x < 0 \\ e^{-\lambda} \sum_{i=0}^{k} \frac{\lambda^i}{i!} & k \le x < k+1 \\ & k = 0, 1, \ldots, \infty \end{cases}$	λ	λ

Continuous random variables:

Distribution	PDF ($f_X(k)$)	CDF ($F_X(x)$)	$E[X]$	$\mathrm{Var}(X)$
Uniform$[a, b]$	$\frac{1}{b-a}, \; a \le x \le b$	$\begin{cases} 0 & x \le a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x \ge b \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exp(λ)	$\lambda e^{-\lambda x}, \; x > 0$	$\begin{cases} 0 & x \le 0 \\ 1-e^{-\lambda x} & x > 0 \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$, $-\infty < x < \infty$	No closed form	μ	σ^2
Gamma(α, β)	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \; x > 0$		$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$ $0 < x < 1$		$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

1. **Markov's inequality:** Let X be a discrete random variable taking non-negative values with a finite mean μ. Then,
$$P(X \ge c) \le \frac{\mu}{c}$$
2. **Chebyshev's inequality:** Let X be a discrete random variable with a finite mean μ and a finite variance σ^2. Then,
$$P(|\, X - \mu \,| \ge k\sigma) \le \frac{1}{k^2}$$

OPTIONS

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A. Useful Data has been mentioned above. ✓ Correct

B. This data attachment is just for a reference & not for an evaluation.

Q2 SA *[3.00 marks]*

Let X be a continuous random variable with PDF as follows:

Find the value of a.

$$f_X(x) = \begin{cases} 2(1-x), & 0 < x < a, \\ 0, & \text{otherwise.} \end{cases}$$

ANSWER

1 (Equal • Numeric)

Q3 SA *[3.00 marks]*

Let X and Y be two independent random variables such that $X \sim \text{Bernoulli}(0.2) \text{ and } Y \sim \text{Bernoulli}(0.4)$. Let another random variable Z be defined as $Z = X + Y$. Find the value of $f_{X|Z=1}(1)$. Enter the answer correct to two decimal places.

ANSWER

0.24 – 0.30 (Range • Numeric)

Q4 SA *[3.00 marks]*

A number from 00 to 99 is selected uniformly at random.The numbers 0 to 9 are treated as two digit numbers and denoted as 00 to 09. Let, • X be the digit in the units place, • Y be the remainder when the number is divided by 4. Find the value of $P(X = 4, Y = 0)$. Enter the answer correct to two decimal places.

ANSWER

0.05 (Equal • Numeric)

Q5 SA *[3.00 marks]*

A random variable X has mean 10 and variance 9. Using Chebyshev's inequality, find a lower bound on $P(4 \leq X \leq 16)$. Enter the answer correct to two decimal places.

ANSWER

0.75 (Equal • Numeric)

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Q6 SA *[3.00 marks]*

A shopping mall issues 400 lucky draw coupons. Among them, one coupon is worth a gift voucher of Rs 2000, one coupon is worth a gift voucher of Rs 1000, and all other coupons are worth Rs 0. If a customer receives 2 coupons drawn at random from the 400, what is the expected value (in Rs) of the prizes won by the customer?

ANSWER

15 (Equal • Numeric)

Q7 MCQ *[3.00 marks]*

CONTEXT / PARAGRAPH

Let X be a discrete random variable with PMF

Based on the above data, answer the given subquestions.

X	0	2	4
$P(X=x)$	1/2	1/3	1/6

Find the value of $E\left[(X-2)^2\right]$.

OPTIONS

A. $\frac{4}{3}$

B. 4

C. $\frac{8}{3}$ ✓ **Correct**

D. 6

Q8 SA *[2.00 marks]*

CONTEXT / PARAGRAPH

Let X be a discrete random variable with PMF

Based on the above data, answer the given subquestions.

X	0	2	4
$P(X=x)$	1/2	1/3	1/6

Using Markov's inequality, obtain an upper bound for $P(X \geq 2)$. Enter the answer correct to two decimal places.

ANSWER

0.64 – 0.70 (Range • Numeric)

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Q9 MCQ *[2.00 marks]*

CONTEXT / PARAGRAPH

Let the random variables $X, Y \sim$ i.i.d Uniform{1, 2, 3, 4}. Based on the above data, answer the given subquestions.

What is the value of $P(\min(X, Y) = 3)$?

OPTIONS

A. $\frac{3}{16}$ ✓ **Correct**

B. $\frac{1}{16}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

Q10 SA *[3.00 marks]*

CONTEXT / PARAGRAPH

Let the random variables $X, Y \sim$ i.i.d Uniform{1, 2, 3, 4}. Based on the above data, answer the given subquestions.

Let $W = \min(X, Y)$, find the value of $P(W \leq 3)$. Enter the answer correct to three decimal places.

ANSWER

0.935 – 0.941 (Range • Numeric)

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Q11 MCQ [3.00 marks]

CONTEXT / PARAGRAPH

Let X_1, X_2, X_3 be i.i.d random variables with the following probability mass function

Based on the above data, answer the given subquestions.

$$f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} k(1+x_1+x_2+x_3), & x_1,x_2,x_3 \in \{0,1\} \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of k.

OPTIONS

A. $\frac{7}{20}$

B. $\frac{1}{20}$ ✓ Correct

C. $\frac{5}{20}$

D. $\frac{9}{20}$

Q12 MCQ [2.00 marks]

CONTEXT / PARAGRAPH

Let X_1, X_2, X_3 be i.i.d random variables with the following probability mass function

Based on the above data, answer the given subquestions.

$$f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} k(1+x_1+x_2+x_3), & x_1,x_2,x_3 \in \{0,1\} \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of $P(X_1 = 1 | X_2 = 1)$.

OPTIONS

A. $\frac{7}{20}$

B. $\frac{12}{20}$

C. $\frac{5}{8}$

D. $\frac{7}{12}$ ✓ Correct

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Q13 SA *[3.00 marks]*

CONTEXT / PARAGRAPH

Let $X_1, X_2, X_3, X_4,$ be i.i.d samples of a discrete random variable X whose PMF is as follows:

Based on the above data, answer the given subquestions.

X	0	1	2	3
$P(X = x)$	1/2	1/4	1/8	1/8

What is the probability that 1 appears exactly once in the samples? Enter the answer correct to two decimal places.

ANSWER

0.39 – 0.45 (Range • Numeric)

Q14 SA *[2.00 marks]*

CONTEXT / PARAGRAPH

Let $X_1, X_2, X_3, X_4,$ be i.i.d samples of a discrete random variable X whose PMF is as follows:

Based on the above data, answer the given subquestions.

X	0	1	2	3
$P(X = x)$	1/2	1/4	1/8	1/8

What is the probability that none of the four samples is equal to 0? Enter the answer correct to two decimal places.

ANSWER

0.03 – 0.09 (Range • Numeric)

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Q15 SA *[1.00 mark]*

CONTEXT / PARAGRAPH

Suppose that the time X is a random variable with probability density

$$f_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{8}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Let U be the event that he event that he has to wait for more than 1 mi

Solution:
According to the question,

$$U = X > 2, \quad \text{and } V = X > 1$$

We know that the conditional probability is given by

$$P(U|V) = \frac{P(U \cap V)}{P(V)} \qquad \ldots(1)$$

Step 1: Calculate numerator

$$\begin{aligned} P(U \cap V) &= P(X > 2 \cap X > 1) \\ &= \underline{\quad \textbf{A} \quad} \\ &= \underline{\quad \textbf{B} \quad} \frac{x}{8} dx \\ &= \underline{\quad \textbf{C} \quad} \end{aligned}$$

Step 2: Calculate denominator

$$\begin{aligned} P(V) &= \int_1^4 \frac{x}{8} dx \\ &= \underline{\quad \textbf{D} \quad} \end{aligned}$$

Step 3: Substituting the values in the equation (1)

$$P(U|V) = \underline{\quad \textbf{E} \quad}$$

Consider the following options for the blanks.

(1) $P(X \ge 2)$
(2) $P(X > 1)$
(3) $\frac{3}{4}$
(4) $\int_2^4$
(5) $\frac{9}{16}$
(6) $\frac{15}{16}$
(7) $\int_1^4$
(8) $\frac{7}{8}$
(9) $\int_0^2$
(10) $P(X > 2)$
(11) $P(X \ge 1)$

Answer the given sub-questions.
Note: Enter only the correct option number (e.g. 8) without any brackets or punctuation.

Enter the correct option number for $\underline{\quad \textbf{A} \quad}$.

ANSWER

10 (Equal • Numeric)

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Q16 SA *[1.00 mark]*

CONTEXT / PARAGRAPH

Suppose that the time X is a random variable with probability density

$$f_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{8}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Let U be the event that he event that he has to wait for more than 1 mi

Solution:
According to the question,

$$U = X > 2, \quad \text{and } V = X > 1$$

We know that the conditional probability is given by

$$P(U|V) = \frac{P(U \cap V)}{P(V)} \qquad \ldots(1)$$

Step 1: Calculate numerator

$$\begin{aligned} P(U \cap V) &= P(X > 2 \cap X > 1) \\ &= \underline{\quad \text{A} \quad} \\ &= \underline{\quad \text{B} \quad} \frac{x}{8} dx \\ &= \underline{\quad \text{C} \quad} \end{aligned}$$

Step 2: Calculate denominator

$$\begin{aligned} P(V) &= \int_1^4 \frac{x}{8} dx \\ &= \underline{\quad \text{D} \quad} \end{aligned}$$

Step 3: Substituting the values in the equation (1)

$$P(U|V) = \underline{\quad \text{E} \quad}$$

Consider the following options for the blanks.

(1) $P(X \ge 2)$
(2) $P(X > 1)$
(3) $\frac{3}{4}$
(4) $\int_2^4$
(5) $\frac{9}{16}$
(6) $\frac{15}{16}$
(7) $\int_1^4$
(8) $\frac{7}{8}$
(9) $\int_0^2$
(10) $P(X > 2)$
(11) $P(X \ge 1)$

Answer the given sub-questions.
Note: Enter only the correct option number (e.g. 8) without any brackets or punctuation.

Enter the correct option number for $\underline{\quad \text{B} \quad}$.

ANSWER

4 (Equal • Numeric)

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Q17 SA *[1.00 mark]*

CONTEXT / PARAGRAPH

Suppose that the time X is a random variable with probability density

$$f_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{8}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Let U be the event that he event that he has to wait for more than 1 mi

Solution:
According to the question,

$$U = X > 2, \quad \text{and } V = X > 1$$

We know that the conditional probability is given by

$$P(U|V) = \frac{P(U \cap V)}{P(V)} \qquad \ldots(1)$$

Step 1: Calculate numerator

$$\begin{aligned} P(U \cap V) &= P(X > 2 \cap X > 1) \\ &= \underline{\quad \text{A} \quad} \\ &= \underline{\quad \text{B} \quad} \frac{x}{8} dx \\ &= \underline{\quad \text{C} \quad} \end{aligned}$$

Step 2: Calculate denominator

$$\begin{aligned} P(V) &= \int_1^4 \frac{x}{8} dx \\ &= \underline{\quad \text{D} \quad} \end{aligned}$$

Step 3: Substituting the values in the equation (1)

$$P(U|V) = \underline{\quad \text{E} \quad}$$

Consider the following options for the blanks.

(1) $P(X \ge 2)$
(2) $P(X > 1)$
(3) $\frac{3}{4}$
(4) $\int_2^4$
(5) $\frac{9}{16}$
(6) $\frac{15}{16}$
(7) $\int_1^4$
(8) $\frac{7}{8}$
(9) $\int_0^2$
(10) $P(X > 2)$
(11) $P(X \ge 1)$

Answer the given sub-questions.
Note: Enter only the correct option number (e.g. 8) without any brackets or punctuation.

Enter the correct option number for $\underline{\quad \text{C} \quad}$.

ANSWER

3 (Equal • Numeric)

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Q18 SA *[1.00 mark]*

CONTEXT / PARAGRAPH

Suppose that the time X is a random variable with probability density

$$f_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{8}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Let U be the event that he event that he has to wait for more than 1 mi

Solution:
According to the question,

$$U = X > 2, \quad \text{and } V = X > 1$$

We know that the conditional probability is given by

$$P(U|V) = \frac{P(U \cap V)}{P(V)} \qquad \ldots(1)$$

Step 1: Calculate numerator

$$\begin{aligned} P(U \cap V) &= P(X > 2 \cap X > 1) \\ &= \underline{\quad \text{A} \quad} \\ &= \underline{\quad \text{B} \quad} \frac{x}{8} dx \\ &= \underline{\quad \text{C} \quad} \end{aligned}$$

Step 2: Calculate denominator

$$\begin{aligned} P(V) &= \int_1^4 \frac{x}{8} dx \\ &= \underline{\quad \text{D} \quad} \end{aligned}$$

Step 3: Substituting the values in the equation (1)

$$P(U|V) = \underline{\quad \text{E} \quad}$$

Consider the following options for the blanks.

(1) $P(X \ge 2)$
(2) $P(X > 1)$
(3) $\frac{3}{4}$
(4) $\int_2^4$
(5) $\frac{9}{16}$
(6) $\frac{15}{16}$
(7) $\int_1^4$
(8) $\frac{7}{8}$
(9) $\int_0^2$
(10) $P(X > 2)$
(11) $P(X \ge 1)$

Answer the given sub-questions.
Note: Enter only the correct option number (e.g. 8) without any brackets or punctuation.

Enter the correct option number for $\underline{\quad \text{D} \quad}$.

ANSWER

6 (Equal • Numeric)

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Q19 SA *[1.00 mark]*

CONTEXT / PARAGRAPH

Suppose that the time X is a random variable with probability density

$$f_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{8}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Let U be the event that he event that he has to wait for more than 1 mi

Solution:
According to the question,

$$U = X > 2, \quad \text{and } V = X > 1$$

We know that the conditional probability is given by

$$P(U|V) = \frac{P(U \cap V)}{P(V)} \qquad \ldots(1)$$

Step 1: Calculate numerator

$$\begin{aligned} P(U \cap V) &= P(X > 2 \cap X > 1) \\ &= \underline{\quad \textbf{A} \quad} \\ &= \underline{\quad \textbf{B} \quad} \frac{x}{8} dx \\ &= \underline{\quad \textbf{C} \quad} \end{aligned}$$

Step 2: Calculate denominator

$$\begin{aligned} P(V) &= \int_1^4 \frac{x}{8} dx \\ &= \underline{\quad \textbf{D} \quad} \end{aligned}$$

Step 3: Substituting the values in the equation (1)

$$P(U|V) = \underline{\quad \textbf{E} \quad}$$

Consider the following options for the blanks.

(1) $P(X \ge 2)$
(2) $P(X > 1)$
(3) $\frac{3}{4}$
(4) $\int_2^4$
(5) $\frac{9}{16}$
(6) $\frac{15}{16}$
(7) $\int_1^4$
(8) $\frac{7}{8}$
(9) $\int_0^2$
(10) $P(X > 2)$
(11) $P(X \ge 1)$

Answer the given sub-questions.
Note: Enter only the correct option number (e.g. 8) without any brackets or punctuation.

Calculate the value for $\underline{\quad \textbf{E} \quad}$. Enter the answer correct to one decimal place.

ANSWER

0.8 (Equal • Numeric)

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